

Introduction to Machine Learning

Week 2

Prof. B. Ravindran, IIT Madras

1. (1 Mark) **State True or False:** Typically, linear regression tend to underperform compared to k-nearest neighbor algorithms when dealing with high-dimensional input spaces.

- (a) True
- (b) False

Sol. b

2. (2 Marks) Given the following dataset, find the uni-variate regression function that best fits the dataset.

X	Y
2	5.5
3	6.5
4	9
10	18.5

- (a) $f(x) = 1 \times x + 4$
- (b) $f(x) = 1 \times x + 5$
- (c) $f(x) = 1.5 \times x + 3$
- (d) $f(x) = 2 \times x + 1$

Soln. C

3. (1 Mark) Given a training data set of 500 instances, with each input instance having 6 dimensions and each output being a scalar value, the dimensions of the design matrix used in applying linear regression to this data is

- (a) 500×6
- (b) 500×7
- (c) 500×6^2
- (d) None of the above

Soln. B

4. (1 Mark) **Assertion A:** Binary encoding is usually preferred over One-hot encoding to represent categorical data (eg. colors, gender etc)

Reason R: Binary encoding is more memory efficient when compared to One-hot encoding

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false

- (d) A is false but R is true

Soln. D

5. (1 Mark) Select the **TRUE** statement.

- (a) Subset selection methods are more likely to improve test error by only focussing on the most important features and by reducing variance in the fit.
- (b) Subset selection methods are more likely to improve train error by only focussing on the most important features and by reducing variance in the fit.
- (c) Subset selection methods are more likely to improve both test and train error by focussing on the most important features and by reducing variance in the fit.
- (d) Subset selection methods don't help in performance gain in any way.

Sol. a

6. (1 Mark) Rank the 3 subset selection methods in terms of computational efficiency:

- (a) Forward stepwise selection, best subset selection, and forward stagewise regression.
- (b) forward stepwise selection, forward stagewise regression and best subset selection.
- (c) Best subset selection, forward stagewise regression and forward stepwise selection.
- (d) Best subset selection, forward stepwise selection and forward stagewise regression.

Sol. b

7. (1 Mark) Choose the **TRUE** statements from the following: (Multiple correct choice)

- (a) Ridge regression since it reduces the coefficients of all variables, makes the final fit a lot more interpretable.
- (b) Lasso regression since it doesn't deal with a squared power is easier to optimize than ridge regression.
- (c) Ridge regression has a more stable optimization than lasso regression.
- (d) Lasso regression is better suited for interpretability than ridge regression.

Sol. c, d

8. (2 Marks) Which of the following statements are TRUE? Let x_i be the i -th datapoint in a dataset of N points. Let v represent the first principal component of the dataset. (Multiple answer questions)

- (a) $v = \arg \max \sum_{i=1}^N (v^T x_i)^2$ s.t. $|v| = 1$
- (b) $v = \arg \min \sum_{i=1}^N (v^T x_i)^2$ s.t. $|v| = 1$
- (c) Scaling at the start of performing PCA is done just for better numerical stability and computational benefits but plays no role in determining the final principal components of a dataset.
- (d) The resultant vectors obtained when performing PCA on a dataset can vary based on the scale of the dataset.

Soln. A and D